

OptiFlow - PRODUCT CASE STUDY

A smart patient-flow system designed to reduce waiting and improve coordination across high-volume eye hospitals.

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THE PROBLEM

Despite having world-class doctors and advanced tracking systems, high-volume eye hospitals suffer from critical physical bottlenecks that cripple patient throughput. First, centralized and manual registration creates massive upfront queues, delaying the start of the clinical pipeline for all incoming patients. Second, the mandatory 20-minute pupil dilation process lacks automated timer management, causing chaotic waiting areas, missed patient callbacks, and idle time for doctors. These specific operational choke points drastically increase the overall turnaround time (TAT) and degrade the experience for elderly patients. Therefore, a decentralized, mobile-first system is required to offload initial registration and intelligently orchestrate dilation queues to maximize hospital efficiency.

The Current Challenge



Sankara Eye Hospital's existing HMS has multiple friction points that slow patient flow and reduce revenue



Manual Registration

Patients wait in queue, fill paper forms, writer manually enters data. Each patient takes 4–6 mins. Bottleneck starts here.



Payment Queues

Patients physically walk to billing counter after every service. Long lines, duplicate trips, and lost time for patients and staff.



Uninformed Patients

Patients anxious & unknowing while waiting for counsellors. Doctor time consumed with repetitive basic Q&A sessions.



Pharmacy Delays

Doctor prescribes meds. Patient walks to pharmacy. Staff manually packs while patient waits. More queues, more delays.

PRODUCT INSIGHT

The bottleneck was never clinical. It was the dead time between steps.

WHAT WE OBSERVED

A 20-min dilation wait and a 25-min counsellor wait sit idle while doctors are free

Every service ends in a walk to billing or pharmacy: 15–20 min per billed service

Hospitals have PCs but patients all carry phones; kiosks cost ₹5L+

WHAT IT REVEALED

▶ Waiting time is wasted time. It can become value-adding time

▶ Patients queue for the payment, not the care

▶ The best infrastructure is the one patients already own

WHAT WE BUILT

▶ AI education chatbot + dilation flag on nurse dashboard

Counsellor session 25 to 10 min

▶ Pay Where You Stand + pre-packed pharmacy orders

Zero queue at billing and pharmacy

▶ BYOD mobile-first flow + parallel paper path for the elderly

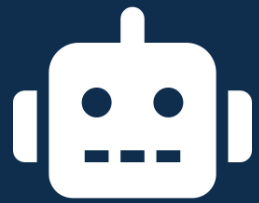
₹0 hardware, no patient excluded

Product principle: a doctor's time is the most valuable resource, so use it only where it is irreplaceable.

Our 4-Module Solution

Seamlessly integrated into your existing HMS — no rip-and-replace required

01



Smart Registration

QR + AI Triage

- ✓ Faster Triage
- ✓ No Paper Forms
- ✓ AI Specialty Match

02



Pay Where You Stand

Mobile-first Billing

- ✓ No Queue Wait
- ✓ Instant Bill
- ✓ Mobile Pay

03



AI Counsellor

Pre-education Bot

- ✓ Reduced Anxiety
- ✓ Doctor Time Saved
- ✓ Better Conversions

04



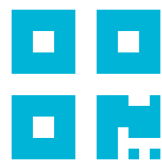
Smart Pharmacy

Pre-pack & Pay

- ✓ Pre-packed Meds
- ✓ Mobile Payment
- ✓ Zero Wait

Each module plugs directly into your existing HMS — activate independently or together

MODULE 01 · Smart Registration & AI Triage



BEFORE — The Old Way

1. Patient enters hospital & joins token queue



2. Waits for token number to be called (10–20 min)

3. Walks to registration desk with paperwork

4. Writer asks questions & manually enters data

5. Writer assigns specialty doctor based on judgment

6. Patient goes to refraction — waits 30+ min for dilation



AFTER — The Smart Way

1. Patient scans QR code at hospital entrance

2. Instant digital token assigned on their phone

3. Fills online form (name, problem checkboxes) on mobile

4. AI (RAG) reads problem & assigns the right specialty doctor

5. AI flags dilation need → shown on nurse screen at refraction

6. Patient called directly — dilation check happens immediately

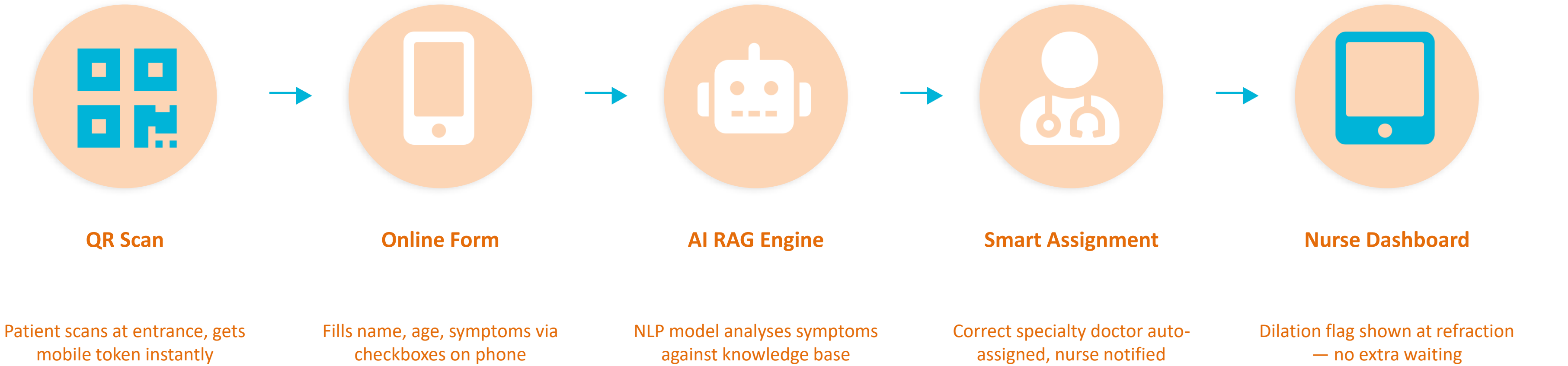


Save 30+ min dilation wait

INNOVATION EDGE

How AI-Powered Triage Works

Module 01 · Registration Flow Deep Dive



Key Benefits of AI Triage

-  No manual data entry — writer freed up
-  Dilation pre-flagged — saves 30 min per patient
-  Accurate specialty assignment — no wrong referrals
-  Zero queues — token & form done on patient's phone

MODULE 02 · Pay Where You Stand



✗ BEFORE — Billing Chaos

1. Doctor writes a scan/test order on paper
2. Nurse manually informs patient to go billing
3. Patient walks to crowded billing counter
4. Waits 10–25 min in queue for billing staff
5. Returns after payment, then waits for scan slot
6. Total extra time: 30–45 min per billed service



✓ AFTER — Pay Where You Stand

1. Doctor/nurse selects scan from HMS — instantly
2. Bill appears on patient's mobile app in real-time
3. Patient pays via UPI/card right from their seat
4. Payment receipt auto-updated in HMS
5. Patient also pays initial registration fee this way
6. Zero trips to billing counter — 100% paperless

💰 Faster payments = improved cash flow & fewer dropouts

INNOVATION EDGE

Mobile-First Billing Flow

Module 02 · How payment happens without leaving your seat

NURSE / DOCTOR SIDE

1. Opens HMS on workstation / tablet
2. Doctor prescribes a scan or procedure
3. Nurse selects the service from dropdown
4. Clicks 'Send Bill to Patient' — done!
5. Sees payment confirmed in real-time
6. Can also initiate pharmacy order here



PATIENT SIDE (Mobile)

1. Receives push notification on phone
2. Opens link — sees itemised bill
3. Pays via UPI, card, net banking
4. Receipt auto-saved in their health record
5. Also pays initial registration fee this way
6. Zero queuing. Zero paper. Instant.

MODULE 03 · AI Pre-Counselling Assistant



✗ BEFORE — Anxious & Uninformed

1. Doctor diagnoses patient, sends them to counsellor
2. Patient waits outside — scared, confused, anxious
3. Has no information about their eye condition
4. When counsellor arrives — floods them with basic Qs
5. Counsellor spends 20–30 min per patient on basics
6. Doctor's time also wasted answering repeated basics



✓ AFTER — Educated & Calm

1. Doctor diagnoses, AI education starts immediately
2. Patient watches videos/reads info about their condition
3. Chatbot answers common questions in real-time
4. Patient arrives at counsellor already educated
5. Counsellor focuses on specifics — session cut to 8–12 min
6. Doctor's precious time saved. Throughput doubled.

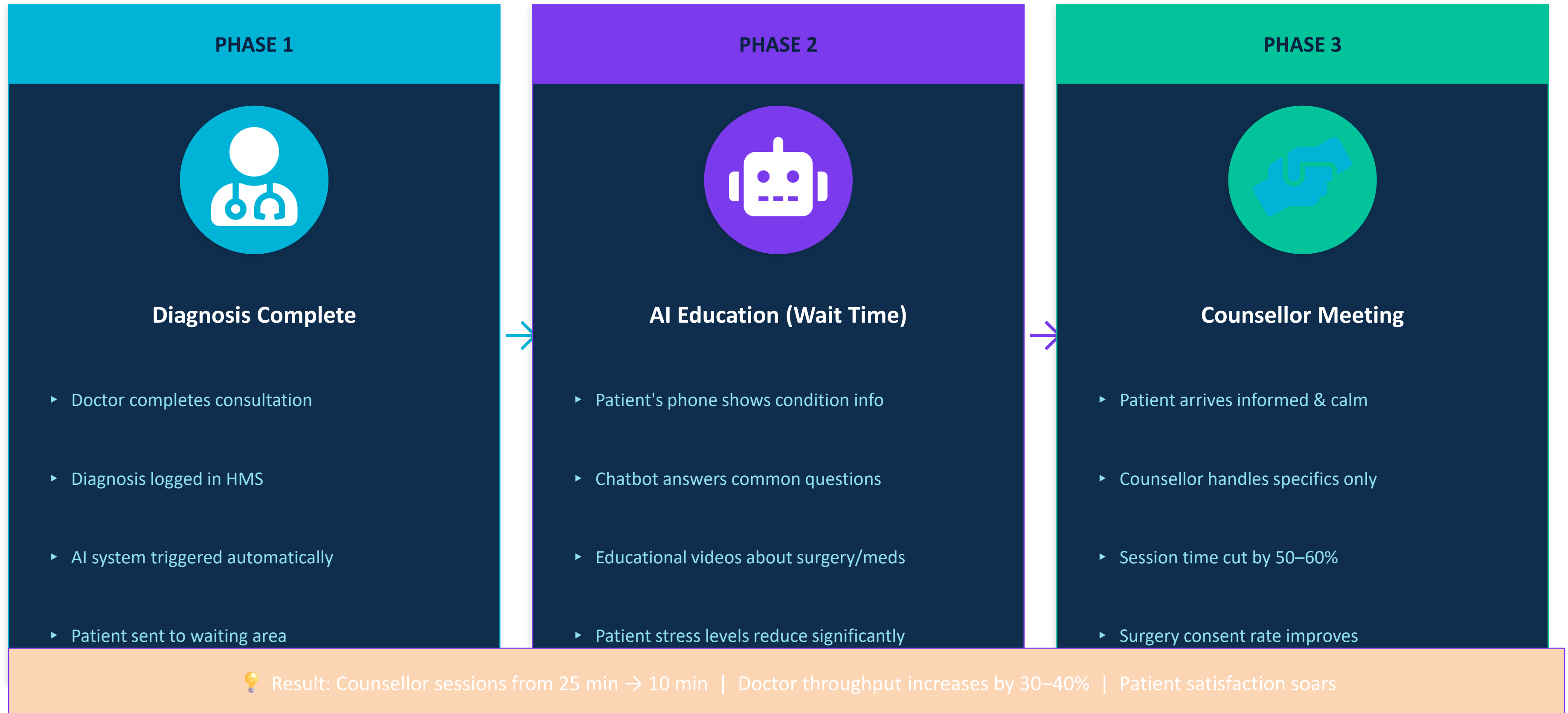


Doctor's time = Most valuable resource. Use it only where irreplaceable.

INNOVATION EDGE

AI Pre-Counselling — Patient Journey

Module 03 · Turning wait time into learning time



MODULE 04 · Smart Pharmacy — Pre-Pack & Pay



✗ BEFORE — Pharmacy Chaos

1. Doctor writes prescription by hand
2. Nurse copies to paper slip for patient
3. Patient walks to pharmacy counter
4. Waits 15–20 min for pharmacist to locate & pack meds
5. Pays in cash — no digital option at pharmacy
6. Total: 25–30 min extra per patient, end of visit



✓ AFTER — Pre-Packed & Ready

1. Nurse selects meds in HMS, confirms with patient
2. Patient confirms: 'Buy here' or 'Buy outside'
3. If here: bill sent to patient mobile via Module 2
4. Pharmacy gets instant prep notice, packs in advance
5. Patient pays via mobile — already done before arrival
6. Patient walks in, picks pre-packed meds, walks out



Pre-packing eliminates wait time & converts cash patients to digital

TECHNICAL FEASIBILITY

Seamless Integration with Existing HMS



No disruption to existing workflows • Modular activation • Fully reversible

PROJECTED IMPACT / TARGETS

Targets based on our initial assumptions; would require validation through a hospital pilot.

35+

min potential saving

Per patient visit

40%

Faster target

Registration processing

60%

Shorter target

Counsellor sessions

Zero

queue target time

At billing & pharmacy

3 Key KPIs:

KPI 1: Patient throughput increase — target 30–40% more patients/day

KPI 2: Billing collection rate — target 95%+ same-day digital payments

KPI 3: Counsellor capacity — target 3x more patients seen per counsellor per day

Societal Impact:

Reduced patient anxiety. Improved access for elderly/non-tech users (parallel paper path).

Model replicable across all 30+ Sankara Eye Hospital locations in India.

COST-EFFECTIVENESS

Zero CapEx, real financial return

₹0

hardware cost

Uses the patient's own smartphone (BYOD) and existing hospital PCs. Traditional kiosks, tablets and servers need ₹5L+.



₹12,00,000+

annual revenue impact per doctor (1 extra patient/hour × ₹500 × 8 hrs)



₹4,80,000

saved yearly in data-entry salaries



₹50L+

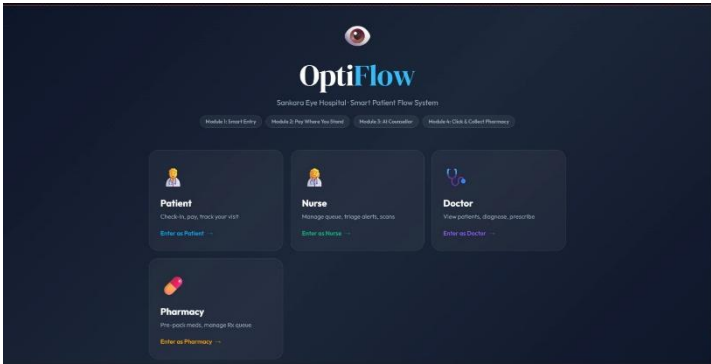
lifetime productivity protected per sight saved, as triage flags retinal detachment

RISKS & MITIGATIONS

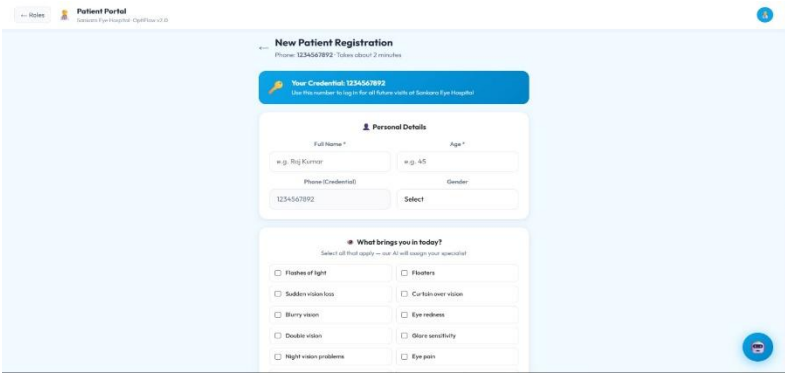
Five risks, each with a built-in safeguard

- | | | |
|---|--|---|
| 1 | Low elderly smartphone adoption | Parallel paper path, QR poster and helper staff |
| 2 | HMS integration complexity | Webhook-based adapter layer, reversible |
| 3 | Patient data privacy | DPDPA/HIPAA aligned, encrypted, no PII in AI training |
| 4 | Staff resistance | One-click nurse dashboard, training, pilot one department |
| 5 | Payment failure / UPI downtime | Cash fallback, retries, offline queuing |

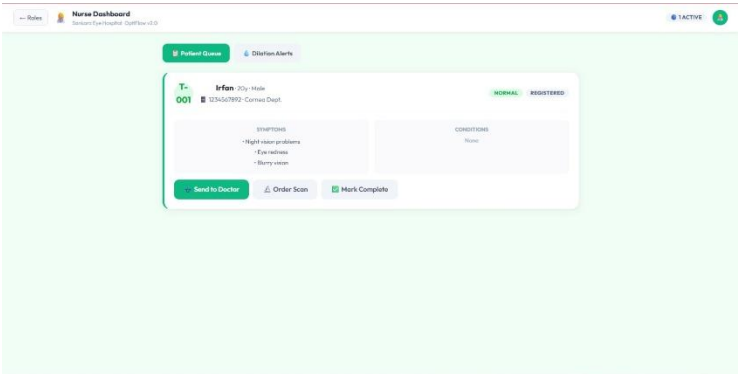
FINAL GOAL



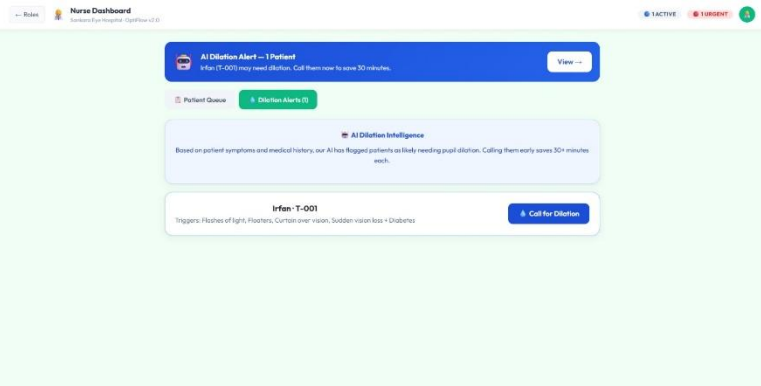
User login



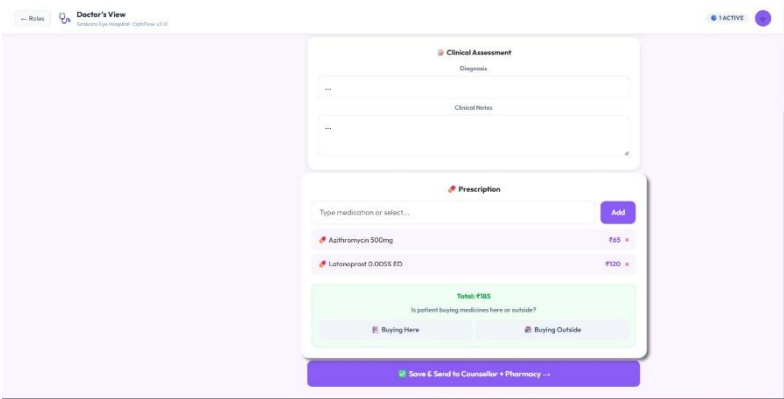
Smart Registration AI,
Assigns Doctor's and
Department from RAG DB



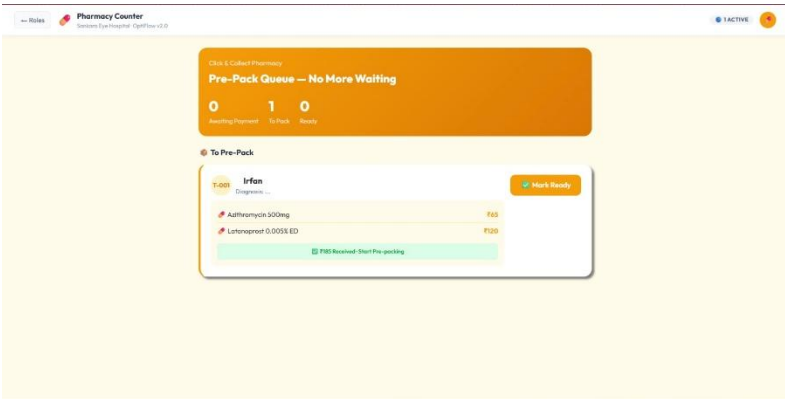
Nurse can select scan so
Patient can do Digital payment
through EHR



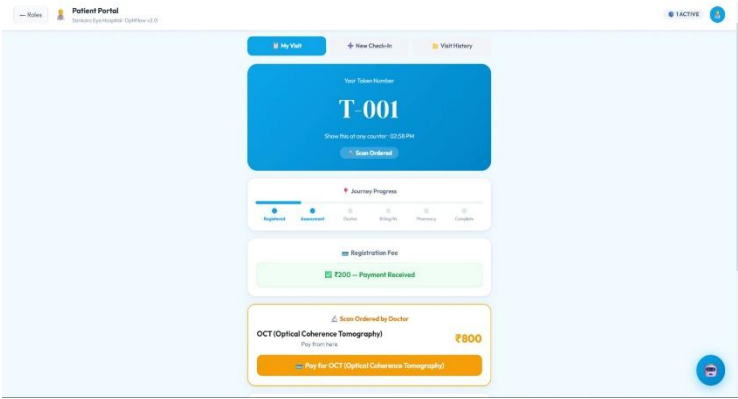
AI gives alert message to the
nurse's dashboard, if the
patient needs dilation, so they
can be called faster



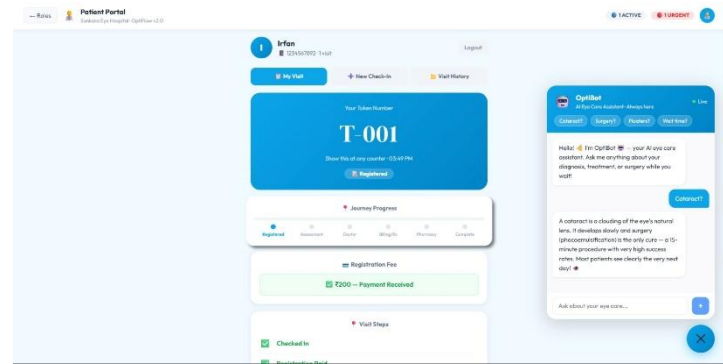
Doctor's can select Meds and
send it to Pharmacy so that
they can pre-pack



Pharmacist can see the bill and
if patient paid online they start
to pre-pack



Patient can pay their registration
fee's, scan bill, pharmacy bill
through this payment gateway



An AI chatbot to educate
patient while they wait for
counselling